Predicting Building Energy Consumption Using Machine Learning

1. Introduction

Energy efficiency has become a critical component of modern infrastructure management. With rising global energy demand and environmental concerns, optimizing building energy consumption is essential. This project explores the application of machine learning techniques to predict building energy consumption based on environmental and operational factors.

2. Problem Statement

The objective of this project is to develop a regression model capable of predicting EnergyConsumption using the following input features: Hour, Temperature, Humidity, SquareFootage, Occupancy, HVACUsage, LightingUsage, and RenewableEnergy.

3. Dataset Description

Feature	Description
Hour	Time of day (0–23)
Temperature	Ambient temperature
Humidity	Relative humidity percentage
SquareFootage	Size of the building
Occupancy	Number of occupants
HVACUsage	HVAC system usage level
LightingUsage	Lighting system usage level
RenewableEnergy	Contribution from renewable sources
EnergyConsumption	Total energy consumed (Target Variable)

4. Data Preprocessing

To ensure consistent feature scaling, standardization was applied to all input features. This process transforms the data using the formula: (X - mean) / standard deviation. Scaling is essential because features such as SquareFootage and Humidity exist on different numerical scales and can disproportionately influence model training.

5. Model Selection and Comparison

Model	MSE	MAE
Linear Regression	70.99	6.68
XGBoost	68.96	6.60
Random Forest	66.34	6.44

Although ensemble models slightly outperformed Linear Regression in terms of error metrics, Linear Regression was selected due to its interpretability, computational efficiency, and ease of deployment.

6. Results and Insights

The final Linear Regression model achieved an MSE of approximately 71 and an MAE of approximately 6.7. This indicates that predictions deviate from actual energy consumption by about 6–7 units on average.

- Temperature and Occupancy significantly impact energy consumption.
- HVAC usage is a major driver during extreme temperature conditions.
- Renewable energy slightly offsets overall energy demand.
- Feature scaling improves model stability and convergence.

7. Future Improvements

- Hyperparameter tuning for ensemble models.
- Feature engineering (interaction terms, polynomial features).
- Time-series modeling for sequential data.
- Deployment as a web or desktop application.
- Integration with real-time IoT sensors.